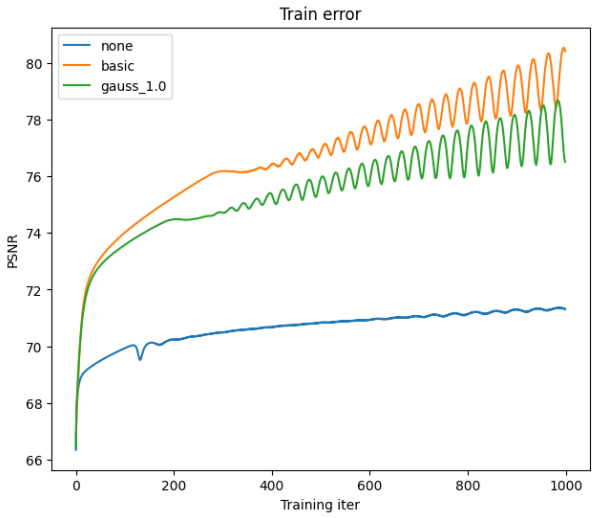
Assignment 2 Report

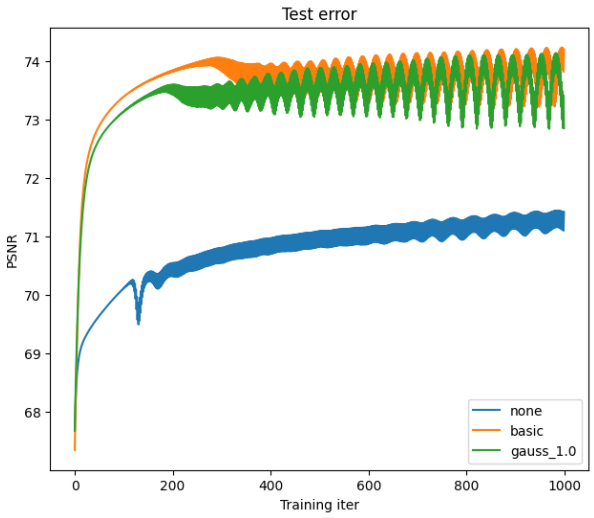
Name(s): Yanjin Li
NetID(s): yanjin2

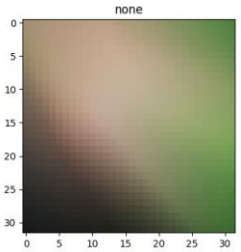
In each the following parts, you should insert the following:

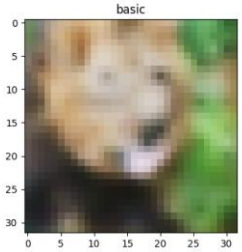
- Train/test loss plots
- Qualitative outputs for GT, No encoding, Basic Positional Encoding, and Fourier Feature Encoding

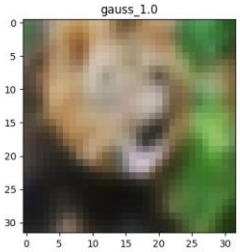
Part 1: Low resolution example

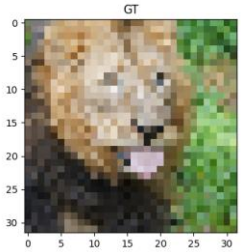




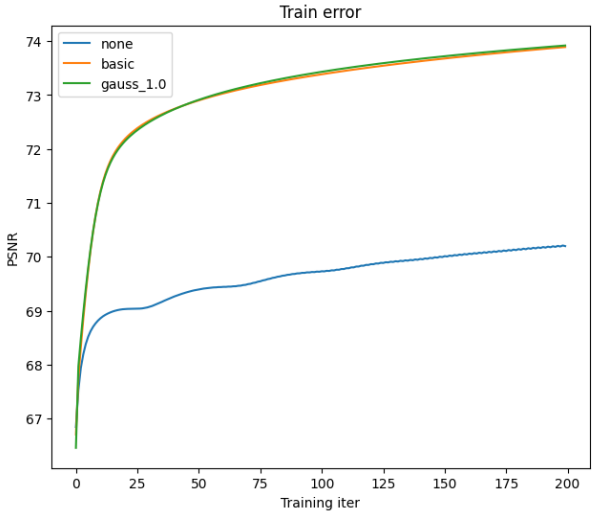


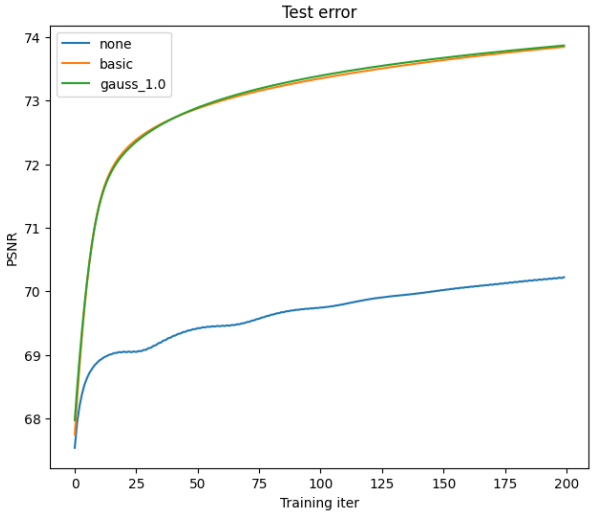


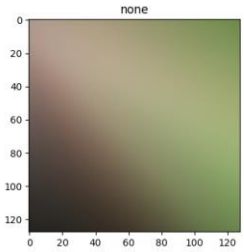


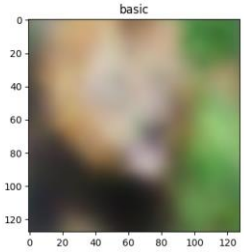


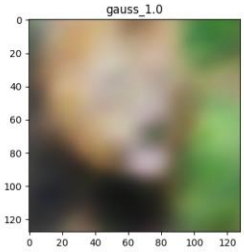
Part 2: High resolution example

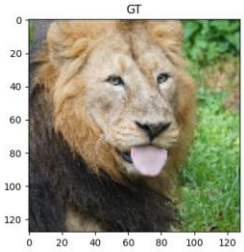




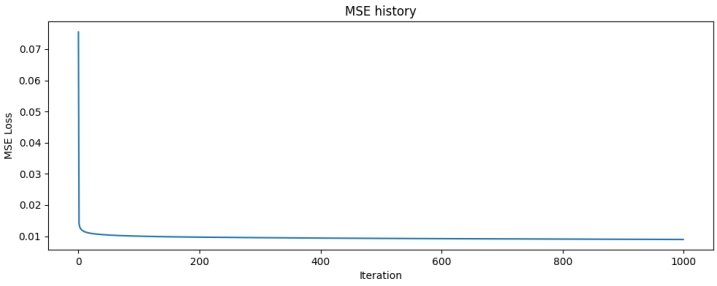


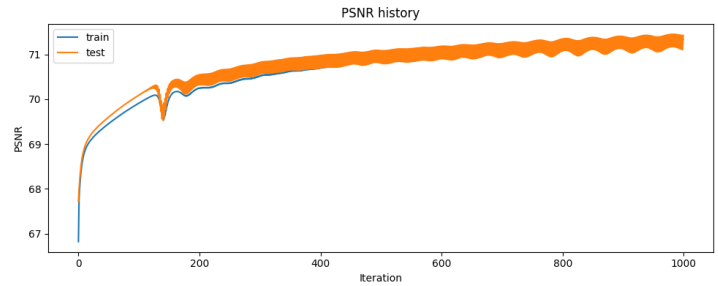


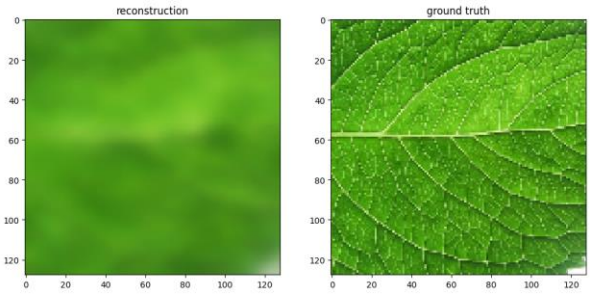




Part 3: High resolution (image of your choice)







(For this part, you can select an image of your choosing and show the performance of your model with the best hyperparameter settings and mapping functions from Part 2. You do not need to show results for all of the mapping functions.)

Part 4: Discussion

Briefly describe the hyperparameter settings you tried and any interesting implementation choices you made.
lr: 1e-3, 1e-2, 5e-2, 1e-1, 5e-1, 0.6, 0.7
hidden_sizes: 128, 256, 512, 1024
num_layers_values: 2, 4, 6, 8
During hyperparameter search, used fewer epochs (200) to speed up experimentation. After identifying promising settings, trained longer (1000 epoches) for final results

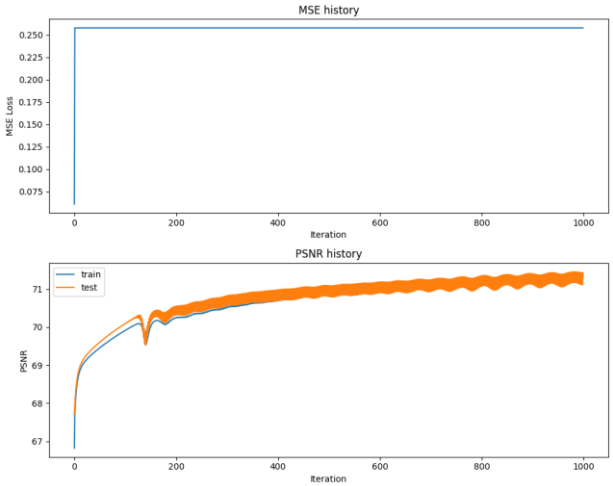
How did the different choices for coordinate mappings functions compare?
None: Lower than other mappings
Basic: Achieves high PSNR relatively quickly, much higher than none
Gauss_1.0: Achieves higher PSNR, but provides limited help

Do you make any interesting observations from the train and test plots?
Both basic and gauss_1.0 converge to higher PSNR. This indicates that adding coordinate encoding helps the network learn details more effectively.

What insights did you gain from your own image example (Part 3)?
Gauss_1.0 is still far from being able to train and recognize detail edges.

Part 5: Extra Credit (optional)

Explain anything you did in detail and provide output images.
Implemented Adam Optimizer


MSE history
MSE Loss
0.250
0.225
0.200
0.175
0.150
0.125
0.100
0.075
0
200
400
600
800
1000
Iteration

PSNR history
train
test
PSNR
71
70
69
68
67
0
200
400
600
800
1000
Iteration